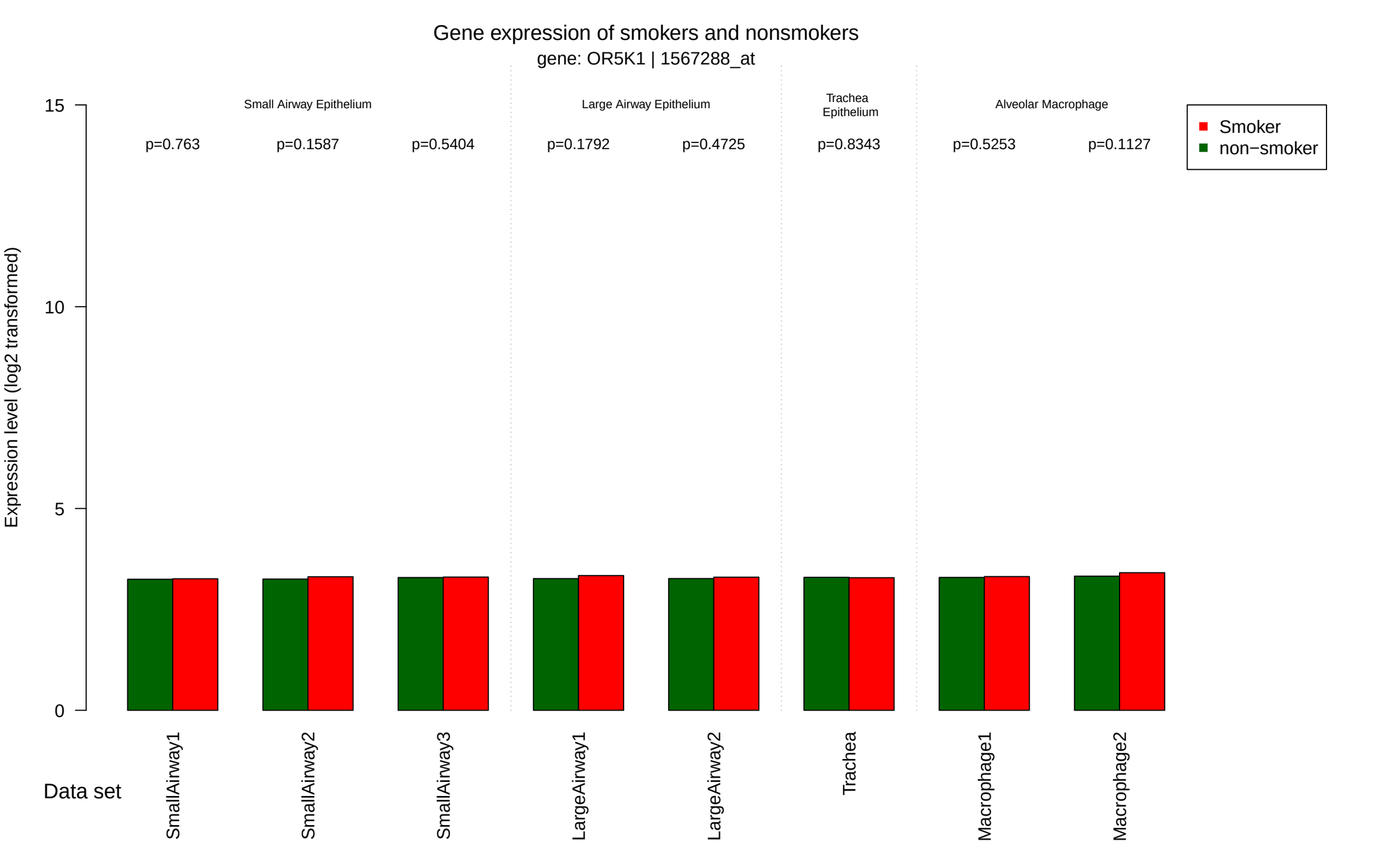
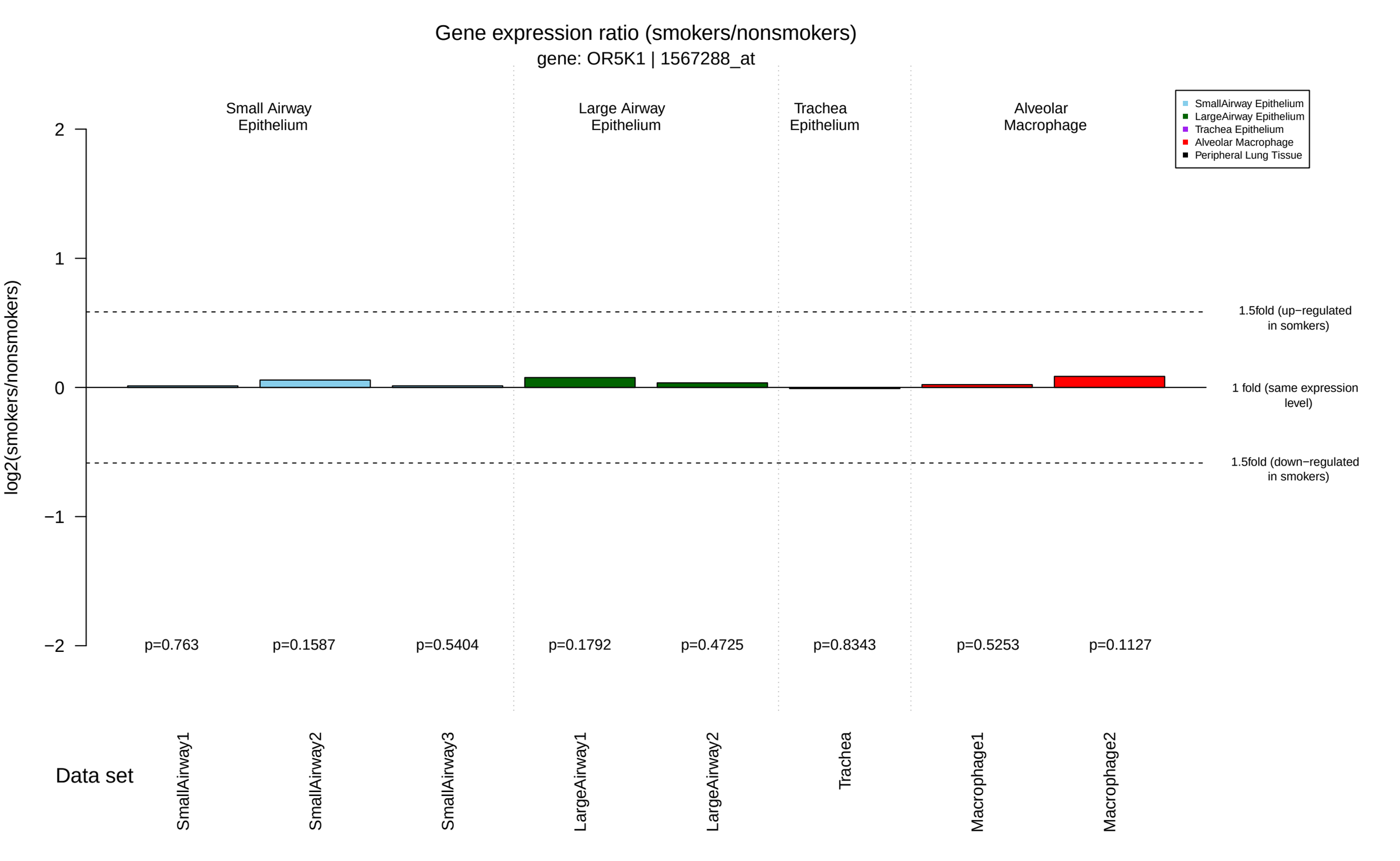
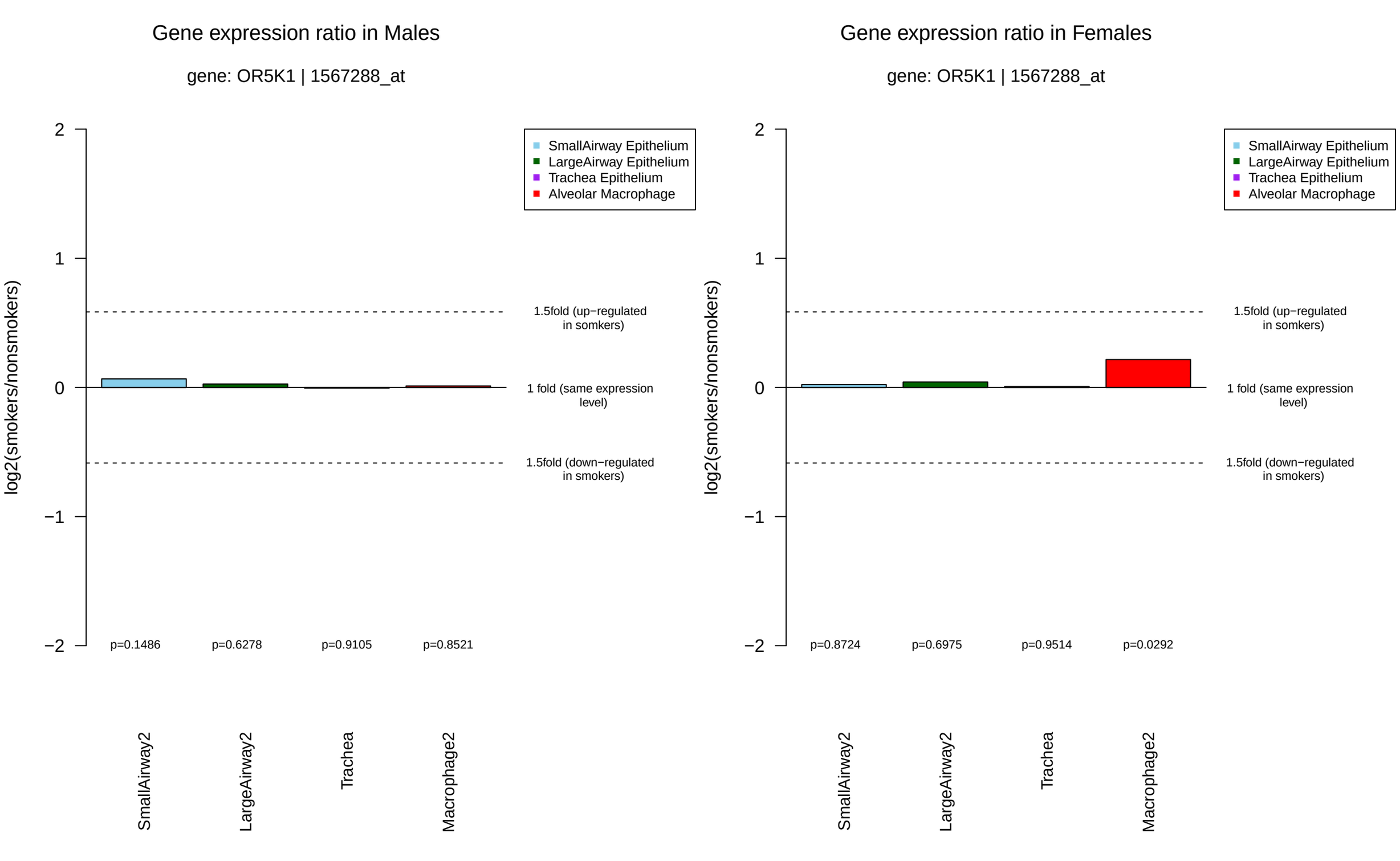
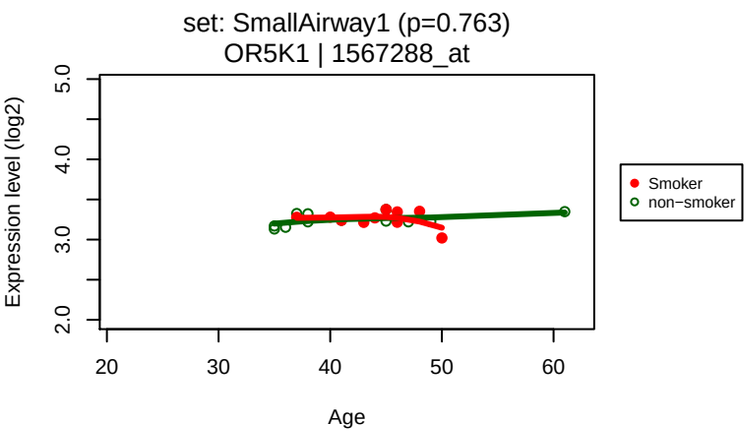
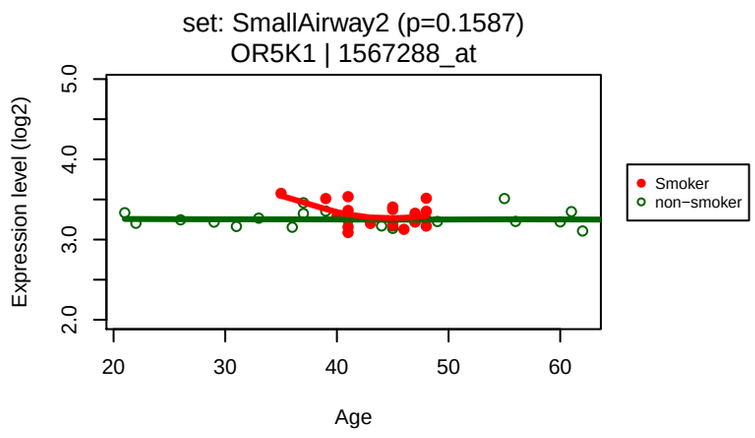
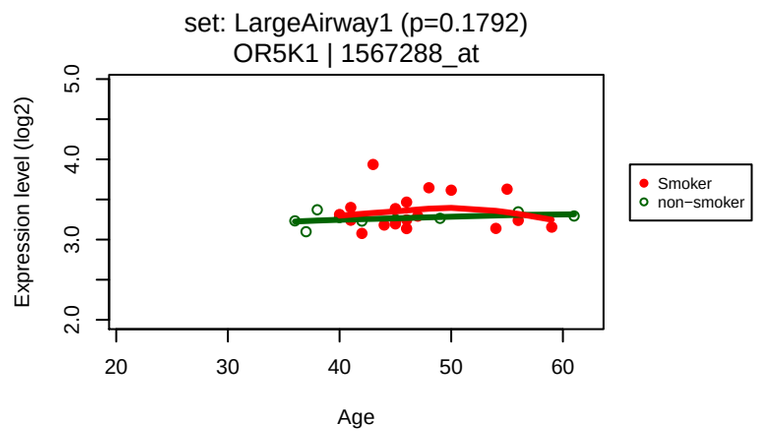

Gene expression of smokers and nonsmokers
gene: OR5K1 | 1567288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.763
p=0.1587
p=0.5404
p=0.1792
p=0.4725
p=0.8343
p=0.5253
p=0.1127
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

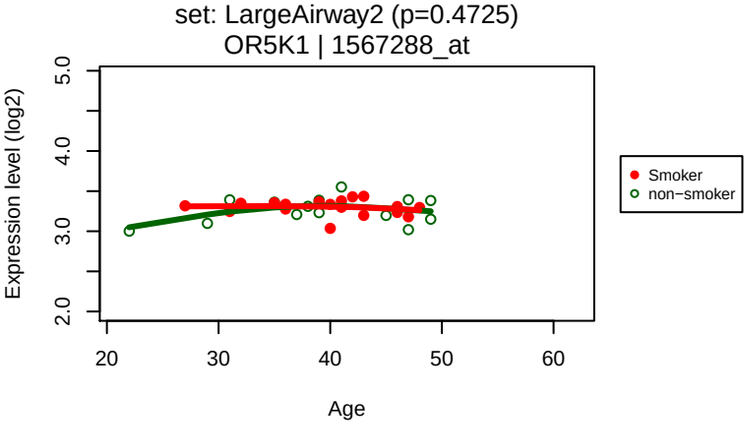

Gene expression ratio (smokers/nonsmokers)
gene: OR5K1 | 1567288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.763
p=0.1587
p=0.5404
p=0.1792
p=0.4725
p=0.8343
p=0.5253
p=0.1127
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

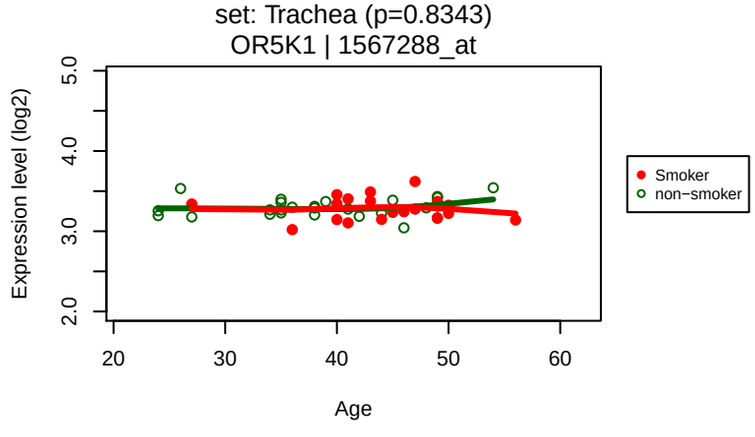

Gene expression ratio in Males
gene: OR5K1 | 1567288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1486
p=0.6278
p=0.9105
p=0.8521
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR5K1 | 1567288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8724
p=0.6975
p=0.9514
p=0.0292
SmallAirway2
LargeAirway2
Trachea
Macrophage2

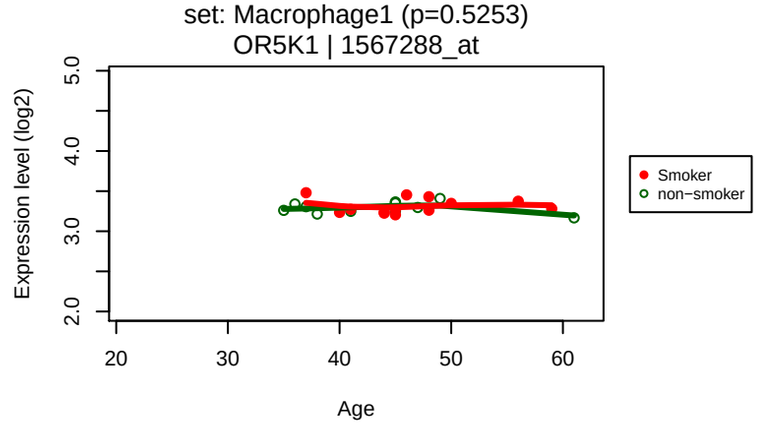

set: SmallAirway1 (p=0.763)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker

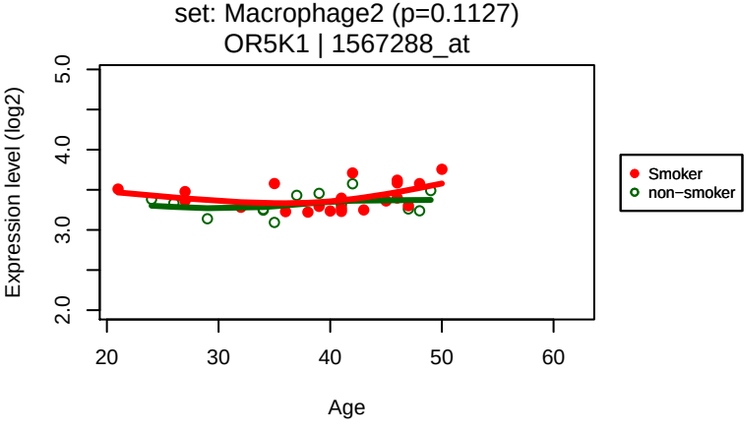

set: SmallAirway2 (p=0.1587)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker

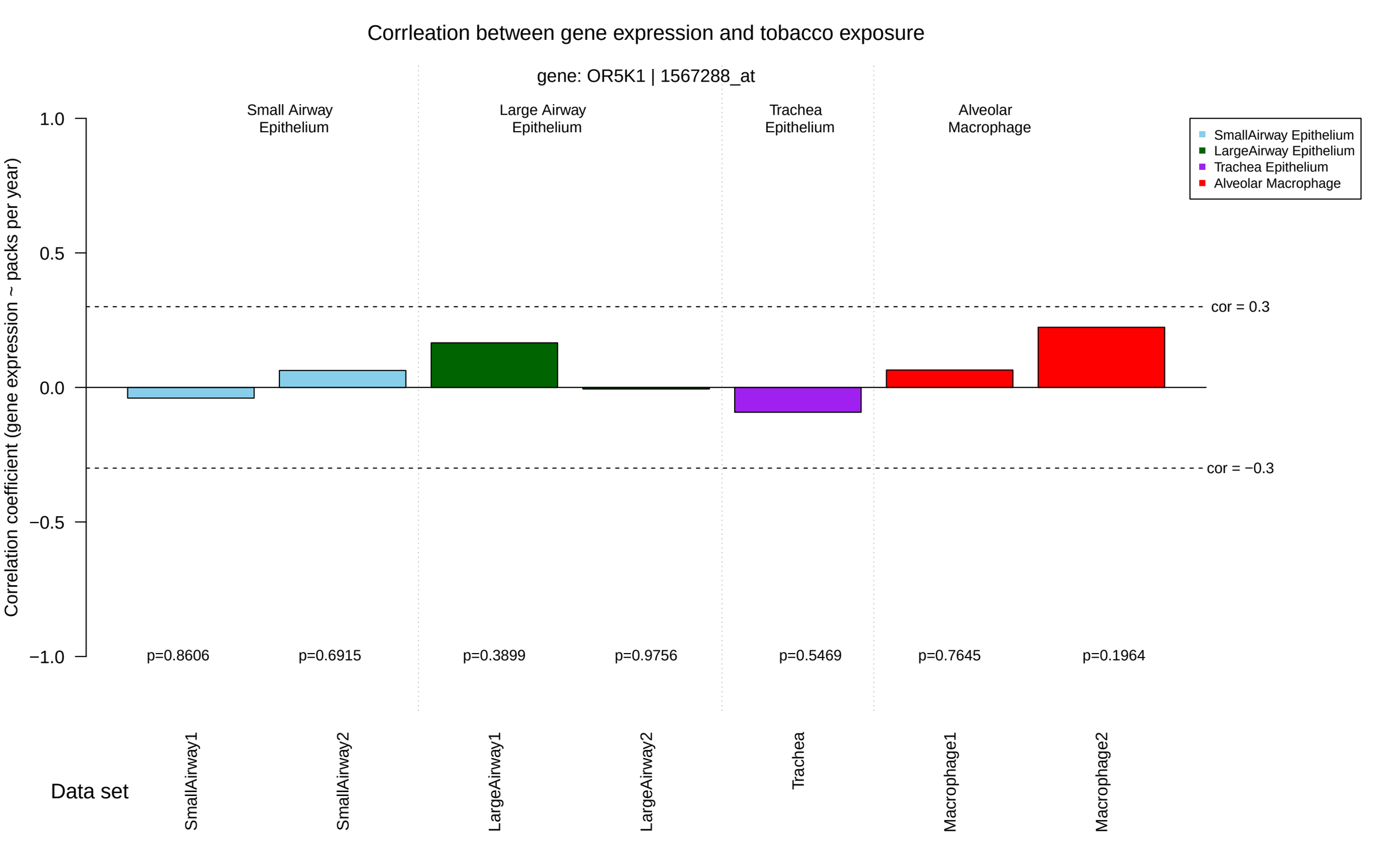

set: LargeAirway1 (p=0.1792)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4725)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8343)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5253)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1127)
OR5K1 | 1567288_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR5K1 | 1567288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8606
p=0.6915
p=0.3899
p=0.9756
p=0.5469
p=0.7645
p=0.1964
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2